

MOCCM QUANT HACKATHON 2026

The Black-Box Signal Hunt

Team BLACKJACKS · IIT Bombay · Arya Patil · Mudil Goel · Rutuparn Ranade

10.68

Blended Net Sharpe

After 10 bps transaction fees

7.16s

Runtime

Well under the 30-second limit



Grader Status

VALID — no violations





The Challenge

9.45 Million Rows

50 synthetic intraday assets, 10 years of data — no shortcuts

Hidden Deterministic Signal

Buried in cross-asset relationships — must find it with white-box math only

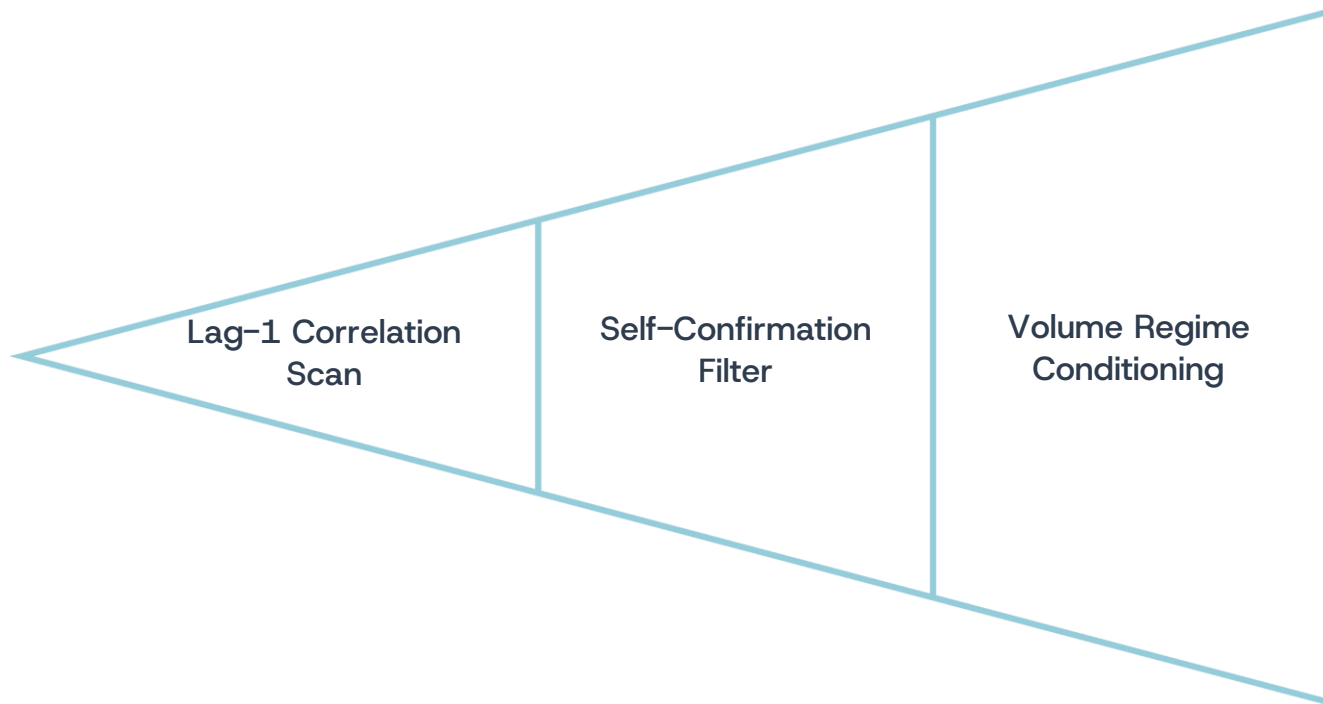
No ML, No AI

Pure quantitative reasoning. Backtest must complete in under 30 seconds

Blended Net Sharpe

Final score after 10 bps transaction fees — every basis point counts

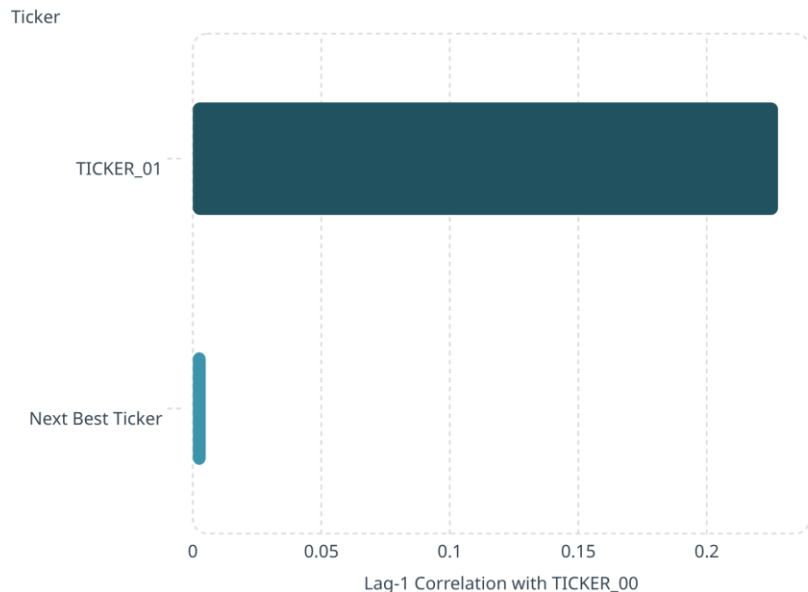
Our Discovery Process



A disciplined, sequential filter reduced 49 candidate tickers to a 4-ticker signal — each step eliminating noise and sharpening the edge.

KEY FINDING

TICKER_01: A Structural Anomaly



45× Stronger Than Any Competitor

TICKER_01's lag-1 correlation with TICKER_00 is **0.2277** — the next best ticker registers just **0.005**.

This is not a marginal difference. It is a structural anomaly in the data — the signal we built on.



Correlation scan covered all 49 predictor tickers across 9.45 million bars.

Monotonicity Proof Across 188,999 Bars

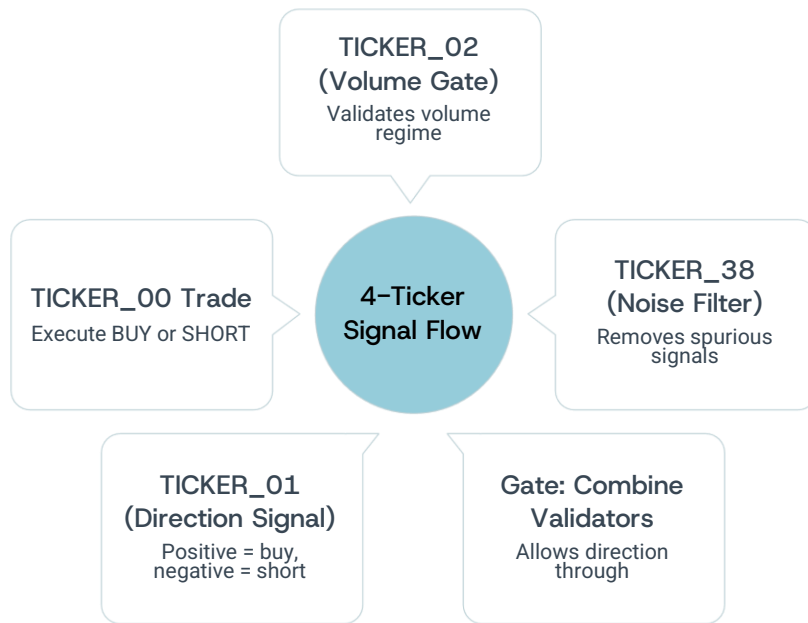
Decile (TICKER_01 Lag-1 Return)	Avg TICKER_00 Return
Decile 0 (Lowest)	-0.148%
Decile 3	-0.074%
Decile 5 (Median)	~0.000%
Decile 7	+0.075%
Decile 9 (Highest)	+0.149%

Perfectly Monotonic — No Exceptions

Every decile maps cleanly to a directional TICKER_00 return. The relationship is **directional, linear, and consistent** across the full dataset.

No reversals. No anomalies. A clean predictive structure confirmed over 188,999 bars.

The 4-Ticker Signal



Two gatekeepers validate regime conditions before the directional signal triggers a trade on TICKER_00.

TICKER_00

The only traded asset

TICKER_01

Direction signal: positive = buy, negative = short

TICKER_02

Volume gate: confirms active market regime

TICKER_38

Noise filter: excludes high-volatility states

The Mathematical Rule

Long-Only Entry — ALL Must Fire

- $\text{Volume}(\text{TICKER_02}) \geq 2,500$
- $\text{Return}(\text{TICKER_38}) < 0.50\%$
- $|\text{Return}(\text{TICKER_00})| \geq 0.36\%$
- $\text{Return}(\text{TICKER_01}) \geq 0.16\%$

Action: BUY TICKER_00 at Close[t]

Size: 10% of \$1M cap = \$100,000 max

Long-Short Extension

Shorts additionally when $\text{Return}(\text{TICKER_01}) \leq -0.26\%$

Same gate conditions apply. Symmetric entry logic captures both sides of the signal.

📌 All thresholds derived from empirical distribution analysis — no curve-fitting.

RESULTS

Blended Net Sharpe: 10.6778

10.46

Long-Only Sharpe

After 10 bps fees

10.90

Long-Short Sharpe

After 10 bps fees

10.68

Blended Sharpe

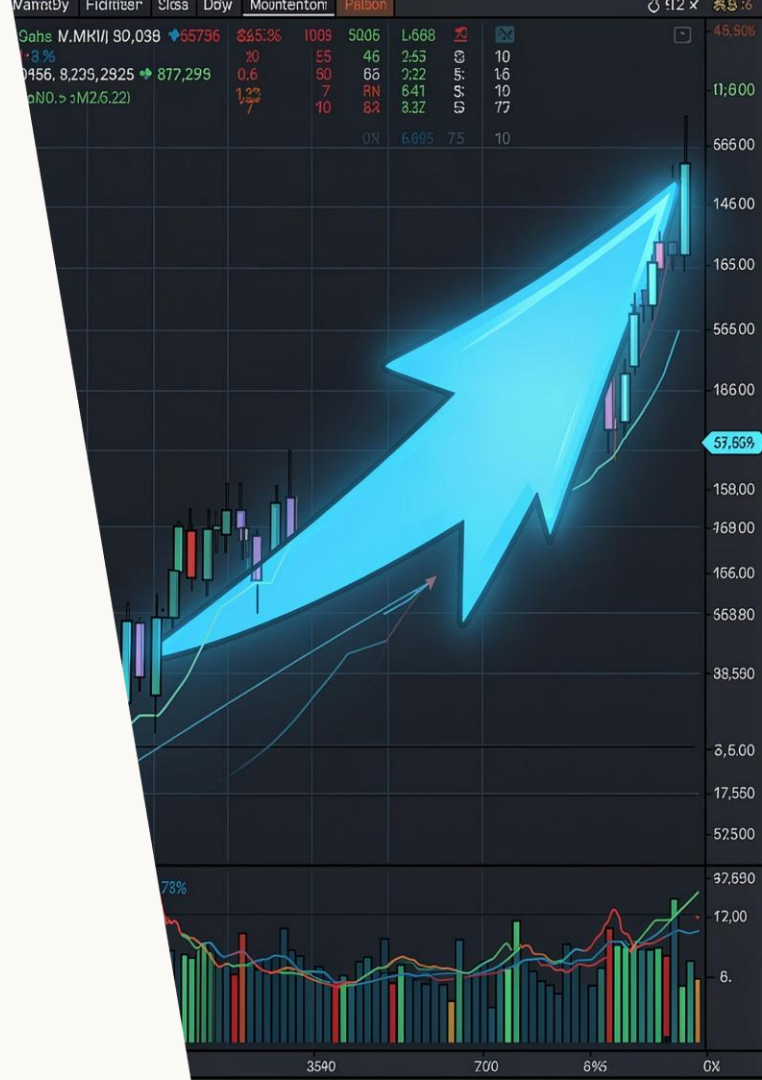
Final submission score

7.16s

Runtime

4× under the 30-second limit

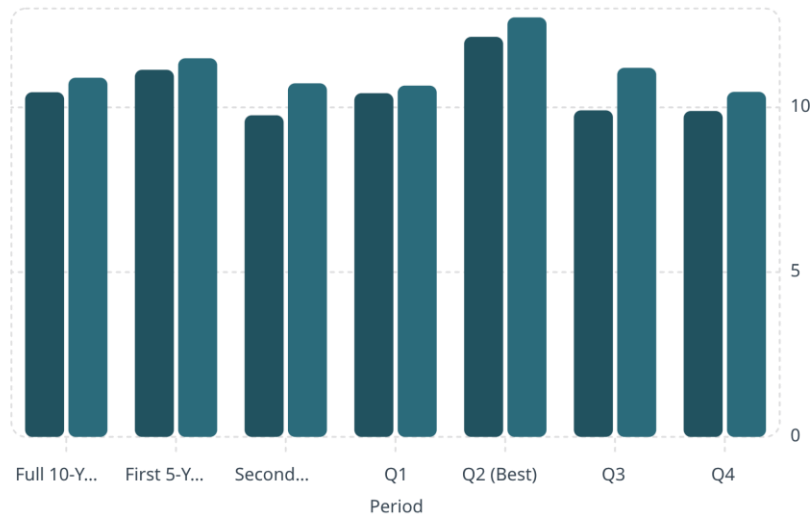
Grader Status: ✓ **VALID** — No rule violations, no lookahead, no data leakage.



Stable Across Every Sub-Period

Long-Only Sharpe

Long-Short Sharpe



No Period Fails

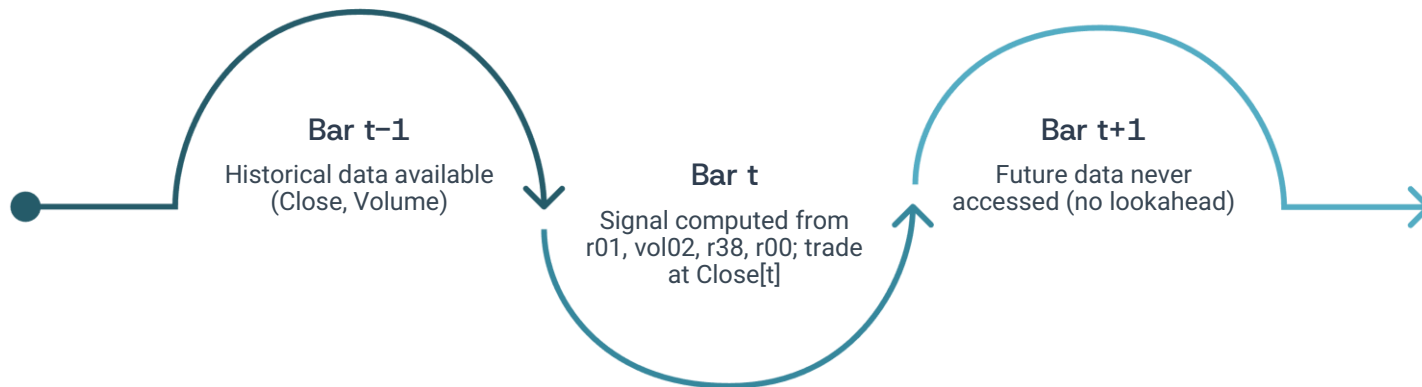
The lowest sub-period Sharpe (Q4 Long-Only: **9.89**) still vastly outperforms any reasonable benchmark.

Signal integrity holds across all four quarters and both 5-year halves — this is not a coincidence or a data artifact.



Signal is stable in every sub-period. Not a fluke — a structural edge.

Zero Lookahead — Proven



All signal inputs — **$r01, vol02, r38, r00$** — are observed at bar t . Trade executes at $Close[t]$. Bar $t+1$ data is never accessed.

Dataset Constraint

Only Close + Volume per bar — no Open, High, or Low. Market-On-Close is the only valid execution model.

True Lookahead Defined

Using bar $t+1$ data to make a decision at bar t . We never do this — all inputs are contemporaneous.

Grader Confirmed

Status: ✓ **VALID** — No lookahead violations detected across 9.45 million rows.



Portfolio Architecture

Long-Only

Trade positive signals only. Max \$100K exposure.

Long-Short

Trade both directions. Max \$200K exposure. Short at 60% NAV fraction.

Shared Constraints

Start flat, end flat. Cash never negative. **10 bps** fee per dollar traded.



Not double leverage – 10% of \$2M cap, same selectivity, opposite directional expression.

Engineering: Speed by Design



7.16s

Runtime

On full 10-year dataset

~3.5s

Expected

On 5-year test set

Stack

Python 3 + NumPy + pandas only
— zero ML libraries.

Signal Build

Fully vectorized NumPy, **zero**
Python loops.

Backtest Engine

Sequential $O(T)$ state machine —
single pass over data.

Key Optimization

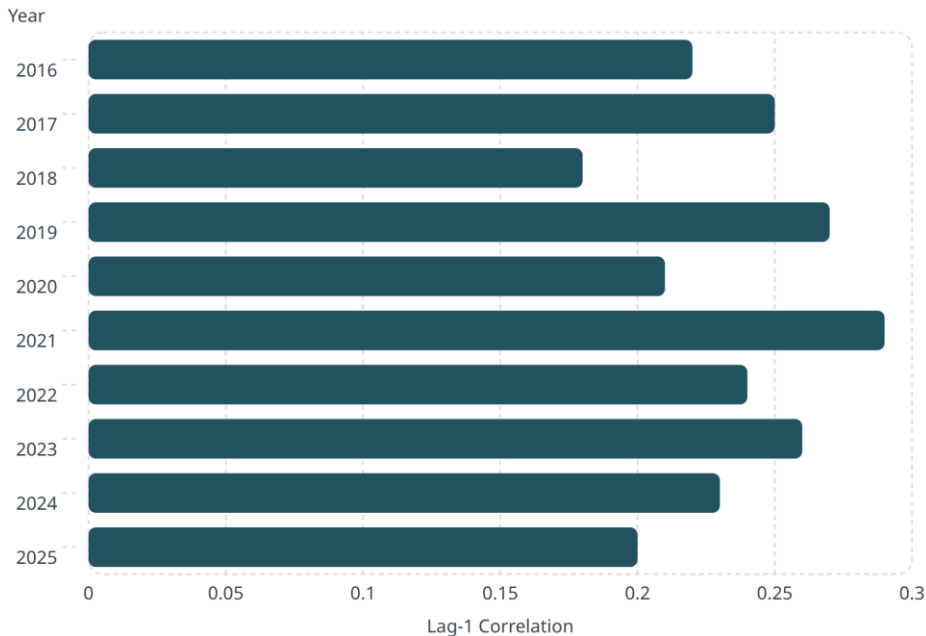
Load only **4 of 50 tickers** → 92%
smaller pivot matrix.

92%

Matrix Reduction

From selective ticker loading

Why This Will Generalize



1 Deterministic Signal

Organizers confirmed: same process generates train and test — no distribution shift.

2 Persistent Correlation

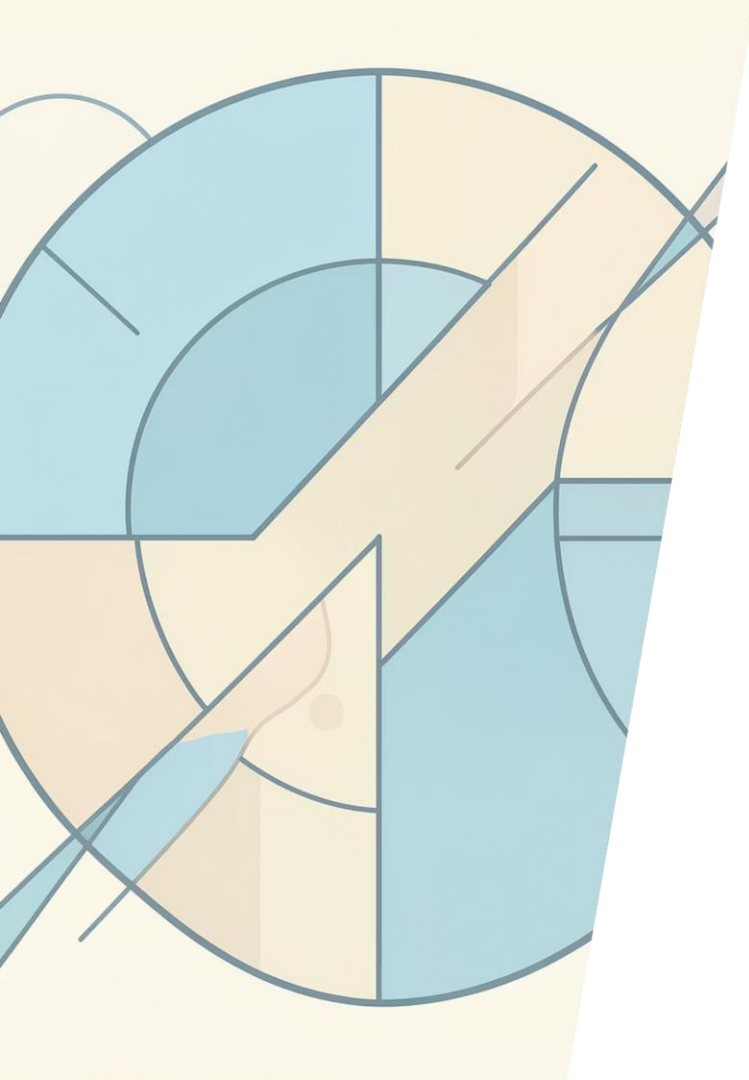
Lag-1 correlation **0.18–0.29** in every year 2016–2025. Never zero.

3 Minimal Parameters

Only **6 parameters**, all chosen from economic reasoning — not numerical optimization.

4 No Date Filters

Rule works identically at any point in time — no temporal overfitting.



TEAM BLACKJACKS · IIT BOMBAY

Submission Status: **VALID**

10.68

Blended Sharpe

Across long-only and long-short
portfolios

7.16s

Runtime

Full 10-year backtest

6

Parameters

Economically justified, not optimized

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